

MODULE 5 NEURONAL INTEGRATION

Gross Anatomy & Neuronal Integration.

GROSS ANATOMY AND NEURONAL INTEGRATION

A reference for the neuronal integration video and lab. This page covers gray and white matter, the names for bundles of fibers and clusters of cell bodies, the reflex arc, types of reflexes, and neuronal pools.

BY THE END

1. Distinguish gray matter from white matter.
2. Define the terms tract, nerve, nucleus, and ganglion.
3. Name the five components of a reflex arc in order.
4. Compare the types of reflexes and describe a neuronal pool.



WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Neuronal integration

drsrennie-stack.github.io/new-build-bio4-solano/neuronal-integration-videos.html

An Overview

This page covers two things: how nervous tissue is organized into the larger structures of the system, and how neurons work together in groups to process information.

THE TERMS THIS PAGE BUILDS ON

Structure	What it is
Gross organization	How nervous tissue is arranged into the brain, cord, and nerves
Neuronal integration	How neurons work together in groups to receive, combine, and respond to information
Gray matter	Regions made mostly of neuron cell bodies, dendrites, and synapses
White matter	Regions made mostly of myelinated axons
Reflex	A fast, automatic, predictable response to a stimulus

Gray Matter and White Matter

Nervous tissue sorts into two looks. Where cell bodies gather, the tissue looks gray; where myelinated axons run, it looks white. Compare them.

GRAY MATTER AND WHITE MATTER COMPARED

Type	What it is made of	Where it is
Gray matter	Neuron cell bodies, dendrites, and unmyelinated fibers	The cortex of the brain and cerebellum, and the inner core of the spinal cord
White matter	Myelinated axons running in bundles	Beneath the cortex of the brain, and the outer part of the spinal cord

Bundles of Fibers and Clusters of Cell Bodies

The same structure has a different name in the CNS and the PNS. Learn the four terms as two pairs: bundles of axons, and clusters of cell bodies.

THE FOUR TERMS, WHERE EACH ONE BELONGS, AND WHAT IT IS

Term	Where	What it is
Tract	CNS	A bundle of axons within the central nervous system
Nerve	PNS	A bundle of axons within the peripheral nervous system
Nucleus	CNS	A cluster of neuron cell bodies within the central nervous system
Ganglion	PNS	A cluster of neuron cell bodies within the peripheral nervous system

HOLD ONTO THIS

Two of the four terms are bundles of axons, tract and nerve, and two are clusters of cell bodies, nucleus and ganglion. Decide which kind of structure you are looking at first, then decide whether you are inside the CNS or outside it.

The Reflex Arc

A reflex is fast because it runs a short, fixed pathway. Every reflex arc has the same five components, in order.

1. **Receptor** detects the stimulus
2. **Sensory neuron** carries the signal from the receptor to the CNS
3. **Integration center** one or more synapses in the CNS where the signal is processed
4. **Motor neuron** carries the command from the CNS to the effector
5. **Effector** the muscle or gland that carries out the response

Types of Reflexes

Reflexes are sorted several ways: by the effector, by the number of synapses, and by whether they are inborn. Compare them.

THE TYPES OF REFLEXES AND WHAT EACH NAME IS BASED ON

Reflex type	Basis of the name	Example
Somatic reflex	The effector is skeletal muscle	The patellar, or knee-jerk, reflex
Autonomic reflex	The effector is smooth muscle, cardiac muscle, or a gland	The pupillary light reflex
Monosynaptic reflex	Just one synapse in the integration center	The stretch reflex
Polysynaptic reflex	Two or more synapses, with interneurons	The withdrawal reflex
Innate reflex	Present from birth, not learned	The blink reflex
Learned reflex	Built through practice and repetition	Catching a suddenly dropped object

Neuronal Pools and Circuits

Neurons rarely act alone. They are wired into groups, the neuronal pools, and the wiring pattern shapes how a signal is handled.

THE POOLS AND THE CIRCUIT PATTERNS

Structure	What it is
Neuronal pool	A functional group of interconnected neurons in the CNS that processes information together
Convergence	A circuit in which many neurons feed their signals onto one neuron
Divergence	A circuit in which one neuron spreads its signal to many neurons
Serial processing	A signal passing along a single straight chain of neurons, as in a reflex arc
Parallel processing	A signal sent along several pathways at the same time
Integration	The combining of all the incoming signals so that a neuron produces a single response

Sensory and Motor Pathways

Information travels the CNS on long tracts: sensory signals climb upward, motor commands descend, and most pathways cross to the opposite side.

RECEPTORS, PATHWAYS, AND CROSSING

Structure	What it is
Sensory receptor	A structure that detects a stimulus and triggers a signal in a sensory neuron
Types of receptors	Named for what they detect: mechanoreceptors, thermoreceptors, photoreceptors, chemoreceptors, and nociceptors for pain
Ascending pathways	The tracts that carry sensory information up the spinal cord to the brain
Descending pathways	The tracts that carry motor commands down from the brain to the body
Decussation	The crossing of a pathway to the opposite side, so each half of the brain serves the opposite half of the body

READING THE WIRING IN THE CLINIC

A tap below the kneecap triggers the **patellar reflex**, and a clinician watches the leg kick. That single test checks the whole reflex arc: the receptor, the sensory and motor neurons, and the spinal cord in between. A missing reflex points to a break somewhere along that pathway.

Because most pathways **decussate**, crossing to the opposite side, the right half of the brain controls and senses the left half of the body. This is why a stroke on one side of the brain causes weakness on the opposite side.

The split into **gray** and **white matter** shows up on a brain or spinal cord scan, and a change in that pattern, gray where white belongs or the reverse, is one of the things a radiologist looks for.